RNN-AFT: Recurrent Neural Networks for Predicting Censored Gap Times in Accelerated Failure Time Models via Gehan-Type Rank Loss

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Abstract

Recurrent event data, in which subjects experience the same type of event multiple times, commonly arise in medicine, insurance, reliability, and economics. The Accelerated Failure Time (AFT) model directly links covariates to event times, providing an intuitive interpretation of how factors accelerate or delay events. Unlike the Cox model, which assumes covariates multiply the hazard, AFT naturally predicts actual event times even when the proportional hazards assumption does not hold. However, traditional AFT models struggle to capture sequential dependencies and nonlinear effects. We introduce RNN-AFT, a deep survival model that extends the AFT paradigm to predict times between repeated events, also known as gap times. It uses Gated Recurrent Units (GRUs), which offer an efficient solution for sequential tasks compared to Long Short-Term Memory (LSTM) networks, known for their complex structure and higher computational cost. To handle censoring, the model employs a Gehan-type rank-based loss function, whose theoretical properties are well established in the literature. Because this loss involves computationally intensive pairwise summations, we adopt a subsampling technique that significantly reduces computational cost while preserving unbiased gradient estimates. Early simulations show that RNN-AFT achieves strong predictive performance, measured by adjusted mean squared error (AMSE) and inverse probability of censoring weighted (IPCW) concordance index, while accurately capturing nonlinear effects and realistic temporal patterns in repeated events. This work constitutes the first deep learning extension of AFT models to recurrent events, providing a flexible and robust framework for analyzing censored recurrent event data.

1 Introduction

Recurrent event data, in which individual subjects may experience the same event multiple times, are common in a variety of domains, including chronic disease progression, insurance claims, equipment failures, and economic cycles. Accurate modeling of these events is crucial for understanding risk patterns, predicting future occurrences, and informing decision-making in both clinical and industrial settings.

Traditional recurrent-event models are often hazard-based, such as extensions of the Cox proportional hazards model. While these models provide interpretable risk estimates, they struggle with high-dimensional covariates, nonlinear relationships, and temporal dependencies inherent in recurrent gap times. AFT models offer an alternative by directly modeling event times, but existing extensions to recurrent events are limited in handling nonlinear predictors and the sequential nature of repeated events.

Deep learning has emerged as a powerful tool for survival analysis, capturing complex covariate interactions and nonlinearities. Methods such as DeepSurv and DeepAFT extend classical survival models using neural networks, but they are primarily designed for single-event settings and may be unstable under high censoring when using inverse-probability-of-censoring-weighted (IPCW) losses. Rank-based approaches, like DeepR-AFT, improve stability but remain restricted to single-event applications.

To address these limitations, we propose RNN-AFT, a recurrent neural network framework that integrates the AFT paradigm with Gated Recurrent Units (GRUs) to model recurrent gap times while capturing nonlinear covariate effects and temporal dependencies. Preliminary simulations demonstrate that the GRU-based model effectively learns inter-event dependencies and nonlinear patterns, and we use a Gehan-type rank-based loss function—to improve prediction accuracy under high censoring.

This work is an extension of AFT models to recurrent event settings using deep neural networks, providing a flexible and powerful framework for modeling complex recurrent event data in the era of artificial intelligence. The remainder of the paper is organized as follows: Section 2 reviews existing methods, Section 3 details the RNN-AFT framework, Section 4 presents simulation studies, Section **??** demonstrates real-data applications, and Section **??** concludes with a discussion of findings and future directions.

2 Background

2.1 The Single-Event Setting

Consider a study with n subjects, where each subject experiences a single event of interest. For subject i, let S_i denote the event time and $\mathbf{X}_i$ denote a vector of covariates associated with subject i, which may be time-fixed or, in extensions, time-dependent.

2.1.1 Classical AFT Formulation

The AFT model is a popular alternative to hazard-based models for analyzing survival and event-time data. Unlike the proportional hazards model, which models the hazard function, the AFT model directly relates the logarithm of event times to covariates, providing an intuitive interpretation in terms of "acceleration" or "deceleration" of event times. The classical AFT model assumes

$$\log(S_i) = \mathbf{X}_i^\top \boldsymbol{\theta} + \epsilon_i,$$

where $\boldsymbol{\theta} \in \mathbb{R}^p$ is a vector of regression coefficients and ϵ_i is a zero-mean random error term with finite variance. In this framework, each coefficient θ_k represents the effect of the

corresponding covariate X_{ik} on the logarithm of the survival time:

$$\frac{\partial \log(S_i)}{\partial X_{ik}} = \theta_k.$$

Equivalently, on the original time scale, a one-unit increase in X_{ik} multiplies the expected survival time by $\exp(\theta_k)$. Values of $\exp(\theta_k)$ greater than one indicate that the covariate prolongs survival, while values less than one indicate shorter survival. This multiplicative interpretation on the time scale makes the AFT model particularly intuitive and useful in medical and reliability applications.

Parametric AFT Models When the error distribution is fully specified (e.g., normal, extreme value, or logistic), maximum likelihood estimation (MLE) can be used to estimate $\boldsymbol{\theta}$ and other distribution parameters. Let f_ϵ and F_ϵ denote the density and cumulative distribution of ϵ_i. For right-censored data with censoring indicator δ_i, the likelihood is

$$L(\boldsymbol{\theta}) = \prod_{i=1}^{n} \left[f_\epsilon(\log(S_i) - \mathbf{X}_i^\top \boldsymbol{\theta})\right]^{\delta_i} \left[1 - F_\epsilon(\log(S_i) - \mathbf{X}_i^\top \boldsymbol{\theta})\right]^{1-\delta_i}.$$

Maximizing $L(\boldsymbol{\theta})$ yields estimates of $\boldsymbol{\theta}$ and, if parametric, the error distribution parameters.

Semiparametric AFT Models When the distribution of ϵ_i is unspecified, semiparametric AFT models can be estimated using rank-based methods. These methods, including Gehan or log-rank-type estimating equations, rely on the ordering of residuals rather than their exact values. Let $R(\boldsymbol{\beta})$ denote the rank-based estimating function; solving $R(\boldsymbol{\beta}) = 0$ provides a consistent estimator of $\boldsymbol{\beta}$. Rank-based inference is robust to misspecification of the error distribution and accommodates right-censored data naturally.

2.1.2 Deep Survival Models

Recent advances in deep learning have inspired a range of models for survival and event-time analysis, extending classical frameworks to handle high-dimensional, nonlinear, and temporally dependent data. One of the earliest approaches, *DeepSurv*, extends the Cox proportional hazards model by replacing the linear log-risk function with a feedforward neural network. This allows the model to capture complex, nonlinear relationships between covariates and the hazard function, while retaining the interpretability of hazard ratios for covariate effects.

Beyond hazard-based approaches, deep learning has also been applied to Accelerated Failure Time (AFT) models. *DeepAFT* replaces the linear predictor of classical AFT models with a neural network, trained using either a Buckley–James imputation-based loss or an inverse-probability-of-censoring-weighted (IPCW) loss. These methods can model nonlinear covariate effects directly on the log-survival or log-gap times. However, IPCW can become unstable under high censoring, motivating alternative loss formulations such as rank-based Gehan losses.

DeepR-AFT further extends these ideas by incorporating recurrent architectures to model sequences of survival times. While initially proposed for single-event settings, DeepR-AFT

demonstrates the potential of using recurrent neural networks (RNNs) to capture temporal dependencies in event sequences. In all these models, the combination of deep learning with survival analysis allows for flexible, data-driven estimation of event-time distributions while accommodating nonlinear covariate effects, complex interactions, and high-dimensional input features.

Our proposed RNN-AFT framework builds on these advances, explicitly modeling sequences of gap times in recurrent events. By using recurrent units such as GRUs or LSTMs, RNN-AFT captures inter-event dependencies and nonlinear effects while maintaining the intuitive interpretation of the AFT model, bridging the gap between classical survival analysis and modern deep learning approaches.

2.2 The Recurrent Event Setting

2.2.1 The AFT Model for Recurrent Gap Time Data

Consider a study involving n individuals who are observed following an initial event and subsequently monitored for recurrences of the same type. Let $i = 1, \ldots, n$ index the subjects and $j = 0, 1, 2, \ldots$ denote the event order, with $j = 0$ corresponding to the initial event. Define T_{ij} as the *gap time* between the $(j-1)$th and the jth events for subject i.

Among regression frameworks for recurrent event data, the AFT model is particularly appealing because it interprets covariate effects as multiplicative shifts in the time scale. Let $\mathbf{X}_i$ be a $p \times 1$ vector of baseline covariates. The model for the logarithm of the gap times is specified as

$$\log(T_{ij}) = \boldsymbol{\theta}_0^\top \mathbf{X}_i + \varepsilon_{ij}, \tag{1}$$

where $\boldsymbol{\beta}_0$ is the true $p \times 1$ vector of regression parameters, and ε_{ij} represents the random error term accounting for unobserved heterogeneity.

For each subject i, define the error vector $\boldsymbol{\varepsilon}_i = (\varepsilon_{i1}, \varepsilon_{i2}, \ldots)^\top$. The error terms are assumed to have identical marginal distributions across gap times, but their correlation structure within a subject is left unspecified. This flexibility allows for complex dependencies among the ε_{ij}—such as autoregressive or unstructured correlation—without imposing restrictive parametric assumptions.

This model specification is weaker than the shared frailty model, which assumes that the errors are conditionally independent and identically distributed given a subject-specific frailty variable. In contrast, model (1) only requires identical marginal distributions, thereby accommodating more realistic dependence patterns that may arise in recurrent event processes.

In practice, recurrent event data are often subject to right censoring due to loss to follow-up or administrative study termination. Let C_i denote the censoring time for subject i, assumed independent of $\{T_{ij} : j \geq 1\}$ conditional on $\mathbf{X}_i$. Let K_i be the number of observed recurrent events for subject i, satisfying

$$\sum_{j=1}^{K_i} T_{ij} \leq C_i < \sum_{j=1}^{K_i+1} T_{ij}. \tag{2}$$

The censoring indicator for the jth gap time is defined as

$$\delta_{ij} = \mathbb{I}\left(\sum_{l=1}^{j} T_{il} \leq C_i\right), \tag{3}$$

where $\mathbb{I}(\cdot)$ denotes the indicator function.

Define the observed gap time as

$$G_{ij} = \begin{cases} T_{ij}, & j = 1, \ldots, K_i, \\ C_i - \sum_{l=1}^{K_i} T_{il}, & j = K_i + 1, \end{cases} \tag{4}$$

and the corresponding logarithm-transformed time as $Y_{ij} = \log(G_{ij})$.

Hence, the observed data for subject i can be expressed as

$$\mathcal{O}_i = \{(G_{ij}, \delta_{ij}) : j = 1, \ldots, K_i + 1;\ \mathbf{X}_i,\ C_i\}. \tag{5}$$

2.2.2 Deep Learning Integration

Integrating deep learning into the AFT framework for recurrent event data presents several conceptual and methodological challenges. In contrast to single-event survival models, recurrent event processes generate sequences of dependent gap times, where the duration between successive events may depend on prior outcomes and covariate information. Modeling such dependencies requires a mechanism that can summarize complex event histories while accommodating both static and time-varying effects—capabilities beyond those of standard linear AFT or feedforward neural network models.

Recurrent neural networks (RNNs), including architectures such as Long Short-Term Memory (LSTM) and Gated Recurrent Unit (GRU) networks, offer a natural solution to this problem. They maintain hidden states that evolve over time, effectively encoding past information into a latent representation that informs future predictions. In this framework, the logarithm of the gap time can be expressed as a nonlinear function of the previous hidden state and current input features:

$$\log G_{ij} = f_{\mathbf{w}}(\mathbf{h}_{i,j-1}, x_{ij}) + \varepsilon_{ij},$$

where $h_{i,j-1}$ denotes the hidden state summarizing prior event history, x_{ij} represents the current input (e.g., the previous gap time and covariates), $f_{\mathbf{w}}$ is a nonlinear mapping parameterized by the RNN weights and output layer, and ε_{ij} is a random error term with unspecified distribution. This formulation generalizes the traditional AFT model by allowing the effect of past events and covariates to vary nonlinearly across time, thereby capturing richer temporal dependencies.

Despite these advantages, extending AFT models in this manner is not trivial. Learning long-range dependencies remains difficult due to issues such as vanishing and exploding gradients, and the interplay between static and dynamic covariates complicates model specification. Furthermore, recurrent event datasets often exhibit high dimensionality and varying event lengths, which can increase computational cost and the risk of overfitting. The presence of right censoring further complicates model training, as not all gap times are fully observed, requiring loss functions that are robust to incomplete information.

The proposed RNN-AFT model is designed to address these challenges. It incorporates static covariates through the initialization of the hidden state and dynamically updates event information through recurrent connections. By combining the interpretability of the AFT structure with the flexibility of deep recurrent networks, this framework provides a principled and scalable approach for modeling complex temporal dependencies in recurrent event data. The next section formalizes this methodology and details its estimation procedures.

3 Methodology

3.1 RNN-AFT Model Formulation

The proposed Recurrent Neural Network Accelerated Failure Time (RNN-AFT) model extends the classical AFT formulation to accommodate recurrent event data by allowing the logarithm of the gap times to depend nonlinearly on both past event history and covariate information. Rather than assuming a fixed linear relationship between covariates and the log-gap time, the RNN-AFT framework learns a dynamic, history-dependent representation through a recurrent neural network that captures temporal dependencies across successive events.

3.1.1 Hidden State Representation

Let $\mathbf{h}_{ij} \in \mathbb{R}^d$ denote the hidden state for subject i after the j-th event, summarizing all relevant information from the event history and covariates up to that point. Two alternative formulations for incorporating covariates into the recurrent structure are considered.

Covariates at Each Time Step. In this formulation, covariates are concatenated with the previous log-gap time and provided as input to the recurrent unit at each step:

$$\mathbf{x}_{ij} = \begin{bmatrix} \log G_{i,j-1} \\ \mathbf{X}_i \end{bmatrix}, \quad \mathbf{h}_{ij} = \mathrm{RNNCell}(\mathbf{h}_{i,j-1}, \mathbf{x}_{ij}), \quad \hat{\mu}_{ij} = \mathbf{w}^\top \mathbf{h}_{ij} + b,$$

where $\mathrm{RNNCell}(\cdot)$ represents a recurrent unit such as a GRU or LSTM, $\mathbf{X}_i$ denotes the static covariate vector, and $\hat{\mu}_{ij}$ is the predicted conditional expectation of $\log G_{ij}$.

Covariates in the Initial Hidden State. Alternatively, covariates can be embedded in the initial hidden state, while subsequent updates depend only on the previous log-gap times:

$$\mathbf{h}_{i0} = \tanh(W_{\text{init}}\mathbf{X}_i + b_{\text{init}}), \quad \mathbf{x}_{ij} = \log G_{i,j-1}, \quad \mathbf{h}_{ij} = \mathrm{RNNCell}(\mathbf{h}_{i,j-1}, \mathbf{x}_{ij}), \quad \hat{\mu}_{ij} = \mathbf{w}^\top \mathbf{h}_{ij} + b.$$

This design separates static and dynamic components: static covariates influence the sequence through the initial hidden representation, while dynamic updates depend solely on the event process. Such separation reduces input dimensionality, avoids redundant inclusion of covariates across time steps, and enhances model generalization, particularly for subjects with long event histories or high-dimensional predictors.

3.1.2 Nonlinear Mapping to Gap Times

Within this framework, the hidden state defines a nonlinear mapping from event history and covariates to the expected log-gap time:

$$\log G_{ij} = f_{\mathbf{w}}(\mathbf{h}_{i,j-1}, \mathbf{x}_{ij}) + \varepsilon_{ij}, \tag{6}$$

where $f_{\mathbf{w}}$ is a flexible, nonlinear function parameterized by the RNN and output layer weights, and ε_{ij} is a random error term capturing unexplained variation. The corresponding calendar time is obtained recursively as

$$S_{ij} = S_{i,j-1} + G_{ij}, \quad S_{i0} = 0.$$

By leaving the distribution of ε_{ij} unspecified, the model adopts a nonparametric AFT perspective, enabling inference and prediction without restrictive distributional assumptions on the gap-time errors. This flexibility allows the RNN-AFT model to capture complex temporal structures, heterogeneous covariate effects, and nonlinear dependencies inherent in recurrent event processes.

3.1.3 Practical Considerations

Embedding static covariates directly into the initial hidden state provides several practical benefits compared to feeding them into the recurrent unit at every time step. This design substantially reduces the input dimensionality at each update, lowering both the computational cost and memory requirements during training. As a result, the model becomes more scalable to larger datasets or settings with numerous covariates.

By separating static and dynamic information, the recurrent unit can concentrate on modeling temporal dependencies driven by the evolving event process rather than repeatedly processing the same static features. This separation enhances the model's ability to capture long-term relationships across event sequences, improving generalization—particularly for subjects with lengthy event histories.

The approach also mitigates the risk of overfitting that may arise from redundant covariate inclusion across time steps. Its efficiency and parsimony make it well-suited for applications involving high-dimensional predictors or sequences with many recurrent events.

Furthermore, the framework naturally extends to accommodate time-varying covariates. In such cases, dynamic covariates observed at each event can be concatenated with the previous log-gap times and fed into the recurrent unit, while static covariates remain embedded in the initial hidden representation. This hybrid design allows the model to capture both baseline subject characteristics and evolving covariate effects, offering a flexible and scalable solution for complex longitudinal processes.

3.2 Network Architecture

The RNN-AFT model adopts a GRU-based recurrent architecture to model the sequence of gap times for each subject. Let G_{ij} denote the j-th observed gap time for subject i. In this formulation, the recurrent input at each step consists only of the previous log-gap time,

while static covariates are embedded in the initial hidden state. The hidden state is then updated recursively based on the event history through the GRU equations:

$$\mathbf{z}_{ij} = \sigma(W_z \mathbf{x}_{ij} + U_z \mathbf{h}_{i,j-1} + \mathbf{b}_z), \tag{7}$$

$$\mathbf{r}_{ij} = \sigma(W_r \mathbf{x}_{ij} + U_r \mathbf{h}_{i,j-1} + \mathbf{b}_r), \tag{8}$$

$$\tilde{\mathbf{h}}_{ij} = \tanh(W_h \mathbf{x}_{ij} + U_h(\mathbf{r}_{ij} \odot \mathbf{h}_{i,j-1}) + \mathbf{b}_h), \tag{9}$$

$$\mathbf{h}_{ij} = (1 - \mathbf{z}_{ij}) \odot \mathbf{h}_{i,j-1} + \mathbf{z}_{ij} \odot \tilde{\mathbf{h}}_{ij}, \tag{10}$$

where $\mathbf{x}_{ij} = \log G_{i,j-1}$, $\mathbf{z}_{ij}$ and $\mathbf{r}_{ij}$ are the update and reset gates, $\tilde{\mathbf{h}}_{ij}$ is the candidate hidden state, $\odot$ denotes element-wise multiplication, and $\sigma(\cdot)$ is the sigmoid activation function.

The predicted log-gap time is obtained from the hidden state via a fully connected output layer:

$$\hat{\mu}_{ij} = \mathbf{w}^\top \mathbf{h}_{ij} + b,$$

where $\mathbf{w}$ and b denote the output layer weights and bias, respectively. For deeper models, multiple GRU layers can be stacked, with hidden states from lower layers serving as inputs to upper layers. Dropout regularization can also be applied between layers to prevent overfitting, especially for long sequences or high-dimensional covariates.

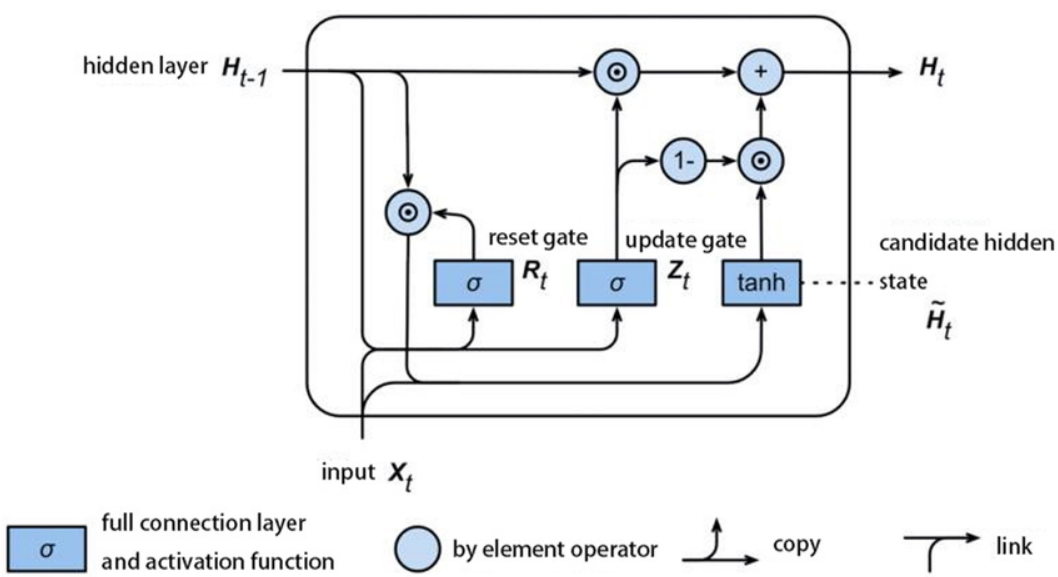


Figure 1: Schematic of the GRU-based RNN-AFT model. Static covariates are embedded in the initial hidden state, while the sequence of log-gap times drives the recurrent updates through GRU layers. The resulting hidden representations are mapped through a fully connected layer to predict the expected log-gap time.

This architecture effectively separates static and dynamic information: the static covariates influence the evolution of the hidden representation through the initial state, while the recurrent unit focuses on modeling temporal dependencies among gap times. This structure enhances generalization, scalability, and interpretability while preserving the flexibility and expressiveness of nonlinear recurrent models.

3.3 Gehan-Type Rank-Based Loss Function

Let

$$e_{ij}(\boldsymbol{\theta}) = \log(G_{ij}) - \boldsymbol{\theta}^\top \mathbf{X}_i$$

denote the residual for the j-th gap time of subject i, where G_{ij} may be observed or censored and $\mathbf{X}_i$ is the baseline covariate vector. Define the counting and at-risk processes on the residual scale as

$$N_{ij}(\boldsymbol{\theta}, t) = \delta_{ij}\, \mathbb{I}\{e_{ij}(\boldsymbol{\theta}) \le t\}, \quad R_{ij}(\boldsymbol{\theta}, t) = \mathbb{I}\{e_{ij}(\boldsymbol{\theta}) \ge t\}.$$

Following the Weighted Risk-Set (WRS) approach [**?**], let K_i be the number of uncensored gap times for subject i and define

$$K_i^* = \max\{K_i, 1\}.$$

If $K_i = 0$, set $G_{i1} = C_i$; otherwise, retain the K_i^* observed gap times. The reduced dataset is

$$\{(G_{ij}, \delta_{ij}) : j = 1, \ldots, K_i^*; \mathbf{X}_i, C_i\},$$

and each gap time is assigned weight $1/K_i^*$ to equalize contributions across subjects.

The *averaged counting and at-risk processes* for subject i are

$$N_i^*(\boldsymbol{\theta}, t) = \frac{1}{K_i^*} \sum_{j=1}^{K_i^*} N_{ij}(\boldsymbol{\theta}, t), \quad R_i^*(\boldsymbol{\theta}, t) = \frac{1}{K_i^*} \sum_{j=1}^{K_i^*} R_{ij}(\boldsymbol{\theta}, t),$$

with corresponding empirical means

$$S_0^*(\boldsymbol{\theta}, t) = \frac{1}{n} \sum_{i=1}^{n} R_i^*(\boldsymbol{\theta}, t), \quad S_1^*(\boldsymbol{\theta}, t) = \frac{1}{n} \sum_{i=1}^{n} \mathbf{X}_i R_i^*(\boldsymbol{\theta}, t).$$

The WRS-based estimating function is then

$$U(\boldsymbol{\theta}) = \sum_{i=1}^{n} \int_{-\infty}^{\infty} w^*(\boldsymbol{\theta}, t) \left[\mathbf{X}_i - \frac{S_1^*(\boldsymbol{\theta}, t)}{S_0^*(\boldsymbol{\theta}, t)} \right] dN_i^*(\boldsymbol{\theta}, t),$$

or equivalently, in discrete form,

$$U(\boldsymbol{\theta}) = \sum_{i=1}^{n} \frac{1}{K_i^*} \sum_{j=1}^{K_i^*} w^*\big(\boldsymbol{\theta}, e_{ij}(\boldsymbol{\theta})\big)\, \delta_{ij} \left[\mathbf{X}_i - \frac{S_1^*\{\boldsymbol{\theta}, e_{ij}(\boldsymbol{\theta})\}}{S_0^*\{\boldsymbol{\theta}, e_{ij}(\boldsymbol{\theta})\}} \right].$$

Under standard regularity conditions, the empirical processes

$$n^{-1} \sum_{i=1}^{n} N_i^*(\boldsymbol{\theta}, t), \quad n^{-1} \sum_{i=1}^{n} \mathbf{X}_i N_i^*(\boldsymbol{\theta}, t), \quad S_0^*(\boldsymbol{\theta}, t), \quad S_1^*(\boldsymbol{\theta}, t)$$

converge uniformly to their population limits and share the same asymptotic distribution as their unweighted counterparts. The map $\boldsymbol{\theta} \mapsto U(\boldsymbol{\theta})$ is compactly differentiable under the supremum norm, ensuring consistency of the solution $\hat{\boldsymbol{\theta}}$ satisfying $U(\hat{\boldsymbol{\theta}}) = 0$.

In single-event settings, it is known that using Gehan's weight produces a monotone estimating function corresponding to a convex objective. Minimizing this objective yields a convex set of solutions, though the minimizer may not be unique. An induced smoothing technique can be applied to the rank-based estimating function with Gehan's weight to ensure smoothness and monotonicity, which is important for stable computation of point estimates and variances (Brown, 2005).

Now, from the WRS-based rank estimating function in Section 3.3, define a Gehan-type weight

$$w^*(\boldsymbol{\theta}, t) = S_0^*(\boldsymbol{\theta}, t),$$

which converges to the standard Gehan weight $w(\boldsymbol{\theta}, t) = S_0(\boldsymbol{\theta}, t)$ for univariate survival data. The Gehan-type estimating function for recurrent gap times is

$$\begin{aligned} U_{\mathrm{G}}(\boldsymbol{\theta}) &= \sum_{i=1}^{n} \int_{-\infty}^{\infty} S_0^*(\boldsymbol{\theta}, t) \left\{ \mathbf{X}_i - \frac{S_1^*(\boldsymbol{\theta}, t)}{S_0^*(\boldsymbol{\theta}, t)} \right\} dN_i^*(\boldsymbol{\theta}, t) \\ &= \sum_{i=1}^{n} \frac{1}{K_i^*} \sum_{j=1}^{K_i^*} S_0^*\big(\boldsymbol{\theta}, e_{ij}(\boldsymbol{\theta})\big) \delta_{ij} \left[\mathbf{X}_i - \frac{S_1^*\big\{\boldsymbol{\theta}, e_{ij}(\boldsymbol{\theta})\big\}}{S_0^*\big\{\boldsymbol{\theta}, e_{ij}(\boldsymbol{\theta})\big\}} \right] \\ &= \frac{1}{n} \sum_{i=1}^{n} \sum_{j=1}^{K_i^*} \sum_{l=1}^{n} \sum_{k=1}^{K_l^*} \frac{\delta_{ij}}{K_i^* K_l^*} \left(\mathbf{X}_i - \mathbf{X}_l\right) \mathbb{I}\Big\{ e_{lk}(\boldsymbol{\theta}) \geq e_{ij}(\boldsymbol{\theta}) \Big\}, \end{aligned}$$

where $e_{ij}(\boldsymbol{\theta}) = \log(G_{ij}) - \boldsymbol{\theta}^\top \mathbf{X}_i$ is the residuals on the log-transformed gap time scale, and the sums iterate over all subjects and their observed or censored gap times. This estimating function is the derivative of the convex objective function

$$L_{\mathrm{G}}(\boldsymbol{\theta}) = \frac{1}{n} \sum_{i=1}^{n} \sum_{j=1}^{K_i^*} \sum_{l=1}^{n} \sum_{k=1}^{K_l^*} \frac{\delta_{ij}}{K_i^* K_l^*} \mathbb{I}\Big[e_{lk}(\boldsymbol{\theta}) - e_{ij}(\boldsymbol{\theta}) \Big]^{-}. \tag{11}$$

Under a nonlinear mapping from the event history and covariates to the expected log-gap time, as in (6), the loss function becomes

$$L_{\mathrm{G}}^{NL}(\mathbf{w}) \propto \sum_{i=1}^{n} \sum_{j=1}^{K_i^*} \sum_{l=1}^{n} \sum_{k=1}^{K_l^*} \frac{\delta_{ij}}{K_i^* K_l^*} \Big[e_{lk}(\mathbf{w}) - e_{ij}(\mathbf{w}) \Big]^{-}, \tag{12}$$

where $[a]^-$ denotes $|\min(a, 0)|$, providing a natural extension of Gehan's rank loss methodology to the nonlinear, nonparametric AFT framework for recurrent gap time data.

3.4 Sub-Sampling Pairs with RNN-AFT

The nonlinear Gehan-type loss $L_{\mathrm{G}}^{NL}(\mathbf{w})$ defined in (12) involves summing over all possible pairs of residuals across subjects and event times, which results in a computational complexity of $O\big((\sum_i K_i^*)^2\big)$. To improve scalability, we adapt a sub-sampling pairs strategy, allowing efficient stochastic gradient updates while preserving the unbiasedness of the gradient estimate.

Let $e_{ij}(\mathbf{w}) = \log(G_{ij}) - \hat{\mu}_{ij}$ denote the residual for the j-th event of subject i, where $\hat{\mu}_{ij}$ is the predicted log-gap time from the RNN-AFT model. The full loss evaluates all pairs (e_{ij}, e_{lk}) such that $\delta_{ij} = 1$ and $(i,j) \neq (l,k)$. Instead of computing all pairs, we randomly select s pairs for each uncensored residual $e_{ij}(\mathbf{w})$ from the set of remaining residuals. Denote the number of uncensored events by $n' = \sum_i \sum_j \delta_{ij}$. The total number of sub-sampled pairs is then $m = n' \times s$.

Training proceeds by forming mini-batches of size b from the m sub-sampled pairs. For a mini-batch containing b pairs (e_{ij}, e_{lk}), the stochastic loss is

$$L^{NL}_{G_{\text{mb}}}(\mathbf{w}) = \sum_{(i,j),(l,k)\in\text{sub-sample}} \frac{\delta_{ij}}{K_i^* K_l^*} \Big[e_{lk}(\mathbf{w}) - e_{ij}(\mathbf{w}) \Big]^{-}.$$

This mini-batch loss is an unbiased estimate of the full Gehan-type loss and allows efficient stochastic gradient descent (SGD) updates of the RNN-AFT parameters $\mathbf{w}$.

Consider a dataset with $n = 3$ subjects, where subject i has K_i^* observed events, and let $s = 2$, $b = 3$. For each uncensored event, we randomly select 2 residuals from all other events. Suppose we obtain the sub-sampled pairs

$$\{(e_{11}, e_{21}), (e_{11}, e_{12}), (e_{21}, e_{32}), (e_{21}, e_{12}), (e_{31}, e_{22}), (e_{31}, e_{11})\}.$$

A mini-batch of size $b = 3$ can then include any 3 of these pairs, with remaining pairs forming subsequent mini-batches within the same epoch.

This sub-sampling procedure significantly reduces computational cost from $O\big((\sum_i K_i^*)^2\big)$ to $O(n's)$ while retaining the stochastic gradient's unbiasedness. The strategy is particularly effective for recurrent event data with long event histories or high-dimensional covariates, and integrates seamlessly with mini-batch SGD for training RNN-AFT models.

Empirically, the choice of s has minimal effect on predictive performance, provided that each uncensored event contributes to multiple sampled pairs. Training stability and convergence are maintained across epochs, and further computational gains can be achieved using GPU acceleration and optimized data loading pipelines.

4 Simulation Study

4.1 Setting

To evaluate the finite-sample performance of the proposed RNN-AFT model, we conducted extensive simulation studies for recurrent gap time data. Let G_{ij} denote the j-th gap time for subject i, with $i = 1, \ldots, n$ and $j = 1, \ldots, K_i$. Conditional on covariates, the log-gap times were generated according to an AFT model:

$$\log G_{ij} \sim f(\mathbf{X}_i) + \epsilon_{ij}, \tag{13}$$

where $\mathbf{X}_i = (x_{i1}, x_{i2}, x_{i3})^\top$ represents baseline covariates, $f(\mathbf{X}_i)$ is the mean function, and ϵ_{ij} are error terms. Covariates were generated as $x_{i1} \sim \text{Bernoulli}(0.5)$, $x_{i2} \sim \mathcal{N}(x_{i1}/3, 1)$, and $x_{i3} \sim \mathcal{N}(x_{i2}/3, 1)$, and three forms of the mean function were considered: a nonlinear function with interactions $f_1(\mathbf{x}) = 3x_1 + 7x_1x_2 - 5x_3$, a GAM-type nonlinear function $f_2(\mathbf{x}) =$

$x_1+x_2^3+e^{0.9x_3}$, and a linear function $f_3(\mathbf{x}) = x_1-6x_2+4x_3$. Error terms were drawn from four standardized distributions—standard normal, Gumbel, and Logistic—all scaled to have zero mean and unit variance. Censoring times C_{ij} were independently sampled from a uniform distribution scaled by a factor τ, i.e.,

$$C_{ij} = \tau \cdot U, \quad U \sim \text{Uniform}(0,1), \tag{14}$$

with τ tuned to achieve approximate censoring percentages of 25%, 50%, and 65%.

To introduce realistic within-subject correlation among repeated gap times, we considered two correlation structures. First, subject-specific random effects (frailty) $b_i \sim \mathcal{N}(0, \sigma_b^2)$ were added to the error terms, so that

$$\epsilon_{ij} = b_i + \eta_{ij}, \quad \eta_{ij} \overset{i.i.d.}{\sim} \text{Error Distribution}, \tag{15}$$

inducing positive correlation among gap times within the same subject. Second, an autoregressive AR(1) structure was considered:

$$\eta_{ij} = \rho\, \eta_{i,j-1} + \sqrt{1-\rho^2}\, \xi_{ij}, \quad \xi_{ij} \overset{i.i.d.}{\sim} \text{Error Distribution}, \tag{16}$$

where $\rho = 0.5$ denotes the lag-1 correlation.

For the main experiments, we fixed $s = 10$ and a mini-batch size of $b = 64$. Optimization was performed using a learning rate of $\alpha = 1 \times 10^{-4}$ and for 2, 3, 5, 10, or 15 epochs to examine the effect of training duration on predictive performance.

Independent training and test datasets were generated for each scenario. Training set sizes included $n_{\text{train}} = 1000$ and $n_{\text{train}} = 5000$, with the number of gap times per subject drawn from a Poisson distribution with mean $\lambda = 2$, while the test set was fixed at $n_{\text{test}} = 2000$. Predictive performance was evaluated using the adjusted mean squared error (AMSE) and the inverse-probability-of-censoring weighted (IPCW) concordance index (C-index). The AMSE accounts for censoring and is defined as

$$\text{AMSE} = \frac{1}{\sum_{i=1}^{n_{\text{test}}} K_i^*} \sum_{i=1}^{n_{\text{test}}} \sum_{j=1}^{K_i^*} \frac{\delta_{ij}}{\widehat{G}(G_{ij})} \big\{ \log(G_{ij}) - \hat{f}(\mathbf{X}_i) \big\}^2, \tag{17}$$

where $\hat{f}(\mathbf{X}_i)$ is the predicted log-gap time from the RNN-AFT model, δ_{ij} is the event indicator, and $\widehat{G}(t)$ is the Kaplan–Meier estimator of the censoring survival function. The IPCW C-index measures the proportion of comparable gap-time pairs whose predicted ordering agrees with the observed ordering:

$$\widehat{C}_{\text{IPCW}} = \frac{\sum_{i \neq k} \sum_{j=1}^{K_i^*} \sum_{l=1}^{K_k^*} \delta_{ij} \frac{\mathbb{I}\{G_{ij} < G_{kl}\}\, \mathbb{I}\{\hat{f}(\mathbf{X}_i) < \hat{f}(\mathbf{X}_k)\}}{\widehat{G}(G_{ij})}}{\sum_{i \neq k} \sum_{j=1}^{K_i^*} \sum_{l=1}^{K_k^*} \delta_{ij} \frac{\mathbb{I}\{G_{ij} < G_{kl}\}}{\widehat{G}(G_{ij})}}, \tag{18}$$

providing a consistent estimate of

$$\mathrm{P}\big(\hat{f}(\mathbf{X}_i) < \hat{f}(\mathbf{X}_k) \mid G_{ij} < G_{kl}\big), \tag{19}$$

which is suitable for assessing predictive accuracy in recurrent event settings.

4.2 Simulation Results

4.2.1 Mean Function Estimation

The results across Tables 1–3 illustrate the robust performance of the RNN-AFT model under diverse data-generating mechanisms, error distributions, censoring levels, and dependence structures. For the nonlinear mean function with interactions from Table 1, the model consistently achieves high IPCW C-indices (typically 0.85–0.94), indicating strong ranking ability even under heavy censoring and within-subject dependence. As expected, AMSE increases with censoring from 25% to 65%, reflecting reduced effective sample size after IPCW adjustment, while the C-index remains relatively stable, highlighting robustness in maintaining relative risk ordering. Performance improves with additional epochs but eventually plateaus or slightly declines under 50–65% censoring for smaller training sets ($n_{\text{train}} = 1000$), indicating mild overfitting; in contrast, larger training sets ($n_{\text{train}} = 5000$) achieve near-optimal performance in just 2–5 epochs, underscoring the RNN's data efficiency and fast convergence. Across error distributions, Gumbel errors generally result in lower AMSE and comparable or higher C-indices, while Logistic errors are more challenging, particularly under heavy censoring. Furthermore, AR(1) dependence consistently yields slightly higher C-indices than frailty-based correlation, suggesting that the RNN effectively captures serial dependence beyond subject-level random effects.

Table 2 presents a more challenging scenario, where the true mean function exhibits smooth nonlinearities in a GAM-type form. Here, the model shows a moderate decline in both C-index and AMSE across all configurations, reflecting increased complexity and the difficulty of learning smooth nonlinear effects. For smaller training sets ($n_{\text{train}} = 1000$), C-indices range from approximately 0.72 to 0.86 under Normal and Gumbel errors, and decrease further under Logistic errors as censoring increases, while AMSE rises steadily. Longer training epochs do not consistently improve performance; in several cases, particularly with 50–65% censoring, extended training leads to declining C-indices despite modest AMSE reductions, indicating overfitting with limited effective sample size. Increasing the training sample size ($n_{\text{train}} = 5000$) stabilizes performance, but improvements plateau beyond 3–5 epochs, emphasizing the importance of early stopping. As in the interaction-based setting, Gumbel errors produce lower AMSE, while Logistic errors remain the most challenging. AR(1) dependence again slightly outperforms frailty-based correlation, demonstrating effective exploitation of temporal dependence.

Finally, Table 3 corresponds to the most favorable scenario, where the true mean function is linear and well aligned with the AFT structure. In this case, the RNN-AFT model attains uniformly high C-indices, typically above 0.90 and frequently exceeding 0.95 under Normal and Gumbel errors, even with 65% censoring. Compared to nonlinear and GAM-type settings, performance is more stable across training epochs, sample sizes, and censoring levels, demonstrating the model's ability to recover linear effects efficiently. While AMSE still increases with heavier censoring, this primarily reflects IPCW variability rather than deterioration in ranking performance. Longer training epochs improve C-index modestly at early stages but lead to substantial AMSE inflation at later epochs, particularly for $n_{\text{train}} = 5000$, highlighting overfitting in the squared-error scale without affecting concordance. Error distribution effects are minimal, with Logistic errors slightly more challenging but still producing

strong C-indices above 0.89. AR(1) dependence continues to confer a modest advantage over frailty-based correlation.

Overall, these results collectively demonstrate the flexibility, robustness, and predictive strength of the RNN-AFT model. It maintains high discriminative performance across linear, interaction-based, and nonlinear GAM-type mean structures, adapts efficiently to varying sample sizes, and is resilient to heavy censoring and complex within-subject dependence. The model is particularly effective in capturing temporal correlations via its recurrent architecture and shows robustness across different error distributions, with Gumbel errors often yielding the most favorable AMSE. In summary, the simulation study confirms that the RNN-AFT framework provides a powerful and generalizable approach for predicting recurrent event times under diverse and realistic scenarios.

Table 1: Simulation results (IPCW C-index / AMSE) for the nonlinear + interaction mean function under different error distributions, censoring levels, and correlation structures.

Error Dist.	Structure	$\%C$	$n_{\text{train}} = 1000$			$n_{\text{train}} = 5000$		
			Epoch 5	Epoch 10	Epoch 15	Epoch 2	Epoch 3	Epoch 5
Normal	Frailty	25	0.849/7.58	0.881/6.52	0.909/5.81	0.879/6.42	0.903/6.35	0.931/7.13
		50	0.863/10.91	0.863/10.64	0.860/13.75	0.855/11.13	0.863/13.40	0.878/18.49
		65	0.891/23.02	0.895/19.80	0.878/18.85	0.899/17.69	0.880/19.56	0.871/23.84
	AR(1)	25	0.851/7.37	0.892/5.92	0.918/6.39	0.882/6.69	0.904/6.17	0.941/6.77
		50	0.874/13.39	0.877/11.86	0.886/13.73	0.864/11.96	0.877/13.09	0.903/18.00
		65	0.901/21.68	0.897/18.10	0.883/18.29	0.903/17.78	0.888/19.27	0.880/24.57
Gumbel	Frailty	25	0.852/8.56	0.887/5.84	0.905/4.77	0.884/6.22	0.905/5.14	0.930/4.14
		50	0.880/11.04	0.892/9.14	0.882/10.83	0.879/9.36	0.865/11.19	0.865/13.97
		65	0.904/20.57	0.902/14.83	0.907/15.27	0.900/14.58	0.900/15.25	0.859/19.72
	AR(1)	25	0.843/8.72	0.895/4.98	0.934/3.08	0.893/5.32	0.912/4.47	0.940/3.56
		50	0.884/9.80	0.899/8.07	0.902/8.24	0.884/8.202	0.887/10.07	0.906/12.42
		65	0.904/19.00	0.919/15.23	0.902/13.39	0.907/13.69	0.906/13.47	0.881/18.58
Logistic	Frailty	25	0.848/10.18	0.875/8.63	0.891/7.62	0.870/9.40	0.887/ 8.74	0.904/8.35
		50	0.878/15.36	0.874/13.42	0.845/15.72	0.863/13.58	0.838/16.08	0.817/19.43
		65	0.893/26.39	0.887/20.94	0.867/21.06	0.870/21.23	0.858/22.36	0.776/27.91
	AR(1)	25	0.933/9.24	0.884/8.04	0.898/7.50	0.882/8.60	0.898/8.39	0.920/8.14
		50	0.886/14.75	0.889/13.03	0.873/15.71	0.878/13.26	0.873/15.70	0.873/18.95
		65	0.898/25.38	0.912/21.07	0.900/20.36	0.893/20.69	0.886/21.50	0.841/27.30

4.2.2 Loss Behavior Across Epochs

To examine the learning dynamics of the RNN-AFT model, we study the evolution of the training loss across epochs under varying data-generating mechanisms. We fix the training sample size at $n_{\text{train}} = 1000$ and the censoring proportion at 25%. For each configuration, the model is trained for 30 epochs, and the training loss is recorded under three error distributions: Normal, Gumbel, and Logistic.

For each mean function—Nonlinear with interactions, GAM-type Nonlinear, and Linear—the results are displayed using a common layout in which rows correspond to error

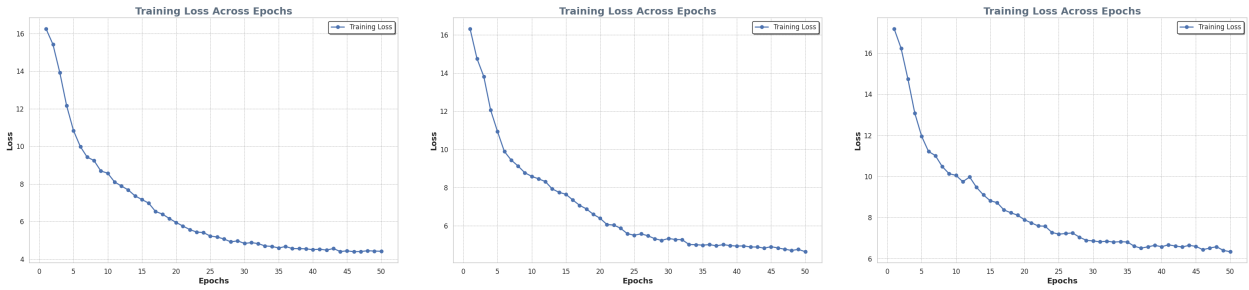
Table 2: Simulation results (IPCW C-index / AMSE) for the GAM-type Nonlinear mean function under different error distributions, censoring levels, and correlation structures.

Error Dist.	Structure	$\%C$	$n_{\text{train}} = 1000$			$n_{\text{train}} = 5000$		
			Epoch 5	Epoch 10	Epoch 15	Epoch 2	Epoch 3	Epoch 5
Normal	Frailty	25	0.841/8.30	0.837/6.57	0.830/5.83	0.836/6.68	0.830/6.80	0.831/10.63
		50	0.827/11.58	0.770/10.28	0.722/9.15	0.794/9.99	0.732/9.61	0.685/12.64
		65	0.849/16.22	0.768/14.16	0.698/13.59	0.806/15.97	0.740/14.40	0.595/13.97
	AR(1)	25	0.856/7.89	0.860/6.08	0.861/5.41	0.860/6.30	0.859/6.408	0.859/9.29
		50	0.853/11.28	0.828/9.46	0.817/8.33	0.841/9.79	0.805/9.54	0.754/11.63
		65	0.874/16.73	0.825/14.72	0.785/13.48	0.855/16.25	0.810/15.59	0.690/14.04
Gumbel	Frailty	25	0.824/7.63	0.816/5.19	0.815/3.89	0.814/4.66	0.804/4.11	0.806/6.71
		50	0.789/8.47	0.735/6.67	0.697/5.40	0.751/6.30	0.689/5.32	0.629/6.48
		65	0.809/10.73	0.740/8.98	0.704/7.73	0.760/9.28	0.714/8.15	0.612/7.24
	AR(1)	25	0.844/7.16	0.848/4.89	0.850/3.55	0.850/4.45	0.847/3.74	0.851/5.57
		50	0.850/8.10	0.827/6.03	0.795/4.75	0.814/6.03	0.776/5.00	0.740/5.89
		65	0.872/9.842	0.810/8.60	0.792/7.17	0.826/8.71	0.797/7.40	0.666/7.43
Logistic	Frailty	25	0.787/7.62	0.774/6.34	0.752/5.92	0.746/6.14	0.726/6.14	0.704/9.66
		50	0.756/9.17	0.686/7.92	0.655/7.45	0.667/8.22	0.619/8.13	0.538/10.44
		65	0.781/11.74	0.689/10.53	0.632/10.09	0.690/11.77	0.607/11.10	0.526/11.83
	AR(1)	25	0.80/6.84	0.812/5.64	0.809/5.35	0.804/5.71	0.791/5.99	0.783/8.86
		50	0.762/7.40	0.756/7.32	0.724/6.93	0.764/7.53	0.699/7.27	0.617/9.20
		65	0.843/11.15	0.769/9.80	0.719/9.91	0.778/10.61	0.729/10.48	0.584/11.25

Table 3: Simulation results (IPCW C-index / AMSE) for the Linear mean function under different error distributions, censoring levels, and correlation structures.

Error Dist.	Structure	$\%C$	$n_{\text{train}} = 1000$			$n_{\text{train}} = 5000$		
			Epoch 5	Epoch 10	Epoch 15	Epoch 2	Epoch 3	Epoch 5
Normal	Frailty	25	0.924/10.51	0.941/14.93	0.944/18.30	0.942/15.64	0.944/19.42	0.943/27.87
		50	0.941/22.94	0.924/25.56	0.897/32.49	0.940/25.78	0.908/34.07	0.902/46.16
		65	0.940/40.89	0.957/34.68	0.934/38.37	0.961/34.23	0.942/39.57	0.877/53.82
	AR(1)	25	0.924/10.79	0.948/14.75	0.95/18.20	0.949/15.18	0.951/19.77	0.951/27.76
		50	0.941/22.91	0.935/26.09	0.924/31.56	0.945/25.09	0.929/32.62	0.927/43.53
		65	0.941/40.26	0.961/33.57	0.942/38.19	0.962/34.07	0.944/39.41	0.902/53.12
Gumbel	Frailty	25	0.917/8.80	0.926/11.85	0.928/16.43	0.933/11.46	0.936/15.16	0.930/22.56
		50	0.935/19.14	0.928/20.13	0.897/28.28	0.931/ 19.15	0.898/27.02	0.885/38.04
		65	0.942/35.76	0.958/28.39	0.940/32.53	0.953/28.73	0.939/32.51	0.858/47.92
	AR(1)	25	0.919/6.23	0.940/7.24	0.940/9.35	0.942/10.86	0.945/14.80	0.945/21.70
		50	0.940/18.93	0.941/19.70	0.919/27.84	0.940/20.25	0.922/26.70	0.907/38.51
		65	0.945/34.85	0.960/28.73	0.943/31.30	0.957/29.05	0.947/31.89	0.886/46.37
Logistic	Frailty	25	0.896/13.33	0.903/17.05	0.889/22.07	0.896/16.71	0.899/21.24	0.892/28.47
		50	0.926/28.12	0.880/30.09	0.825/37.16	0.891/30.09	0.848/37.23	0.811/48.57
		65	0.933/45.16	0.946/38.64	0.890/44.48	0.940/38.61	0.911/42.29	0.793/58.09
	AR(1)	25	0.899/13.94	0.914/18.47	0.919/21.74	0.916/17.79	0.918/22.82	0.916/28.80
		50	0.922/27.75	0.903/29.87	0.882/35.30	0.914/28.75	0.876/37.86	0.855/48.11
		65	0.933/44.19	0.947/39.21	0.899/45.48	0.944/38.13	0.925/43.67	0.836/58.00

distributions. This organization facilitates direct comparison of convergence behavior across distributions within each modeling scenario.



Normal errors Gumbel errors Logistic errors

Figure 2: Training loss across epochs for the Nonlinear + Interaction mean function under different error distributions, with $n_{\text{train}} = 1000$ and 25% censoring.

Normal errors Gumbel errors Logistic errors

Figure 3: Training loss across epochs for the GAM-type Nonlinear mean function under different error distributions, with $n_{\text{train}} = 1000$ and 25% censoring.

Across all three mean functions, the training loss exhibits a stable and generally monotone decrease over epochs, indicating well-behaved optimization and effective learning.

Comparing mean structures, the Linear mean function converges most rapidly and attains the lowest loss levels, reflecting its relative simplicity. The GAM-type Nonlinear mean function shows slightly slower convergence, while the Nonlinear + Interaction mean function exhibits higher initial loss and slower early-stage convergence, consistent with its increased structural complexity. Differences across error distributions are modest overall, though heavier-tailed distributions such as Gumbel and Logistic tend to produce higher initial losses and marginally slower convergence, particularly in the nonlinear settings.

Overall, these results demonstrate that the RNN-AFT model trains robustly across a range of mean functions and error distributions, with no evidence of instability or divergence for large sample sizes.

4.2.3 Model Performance Across Sub-sampling Configurations

To examine the sensitivity of the RNN-AFT model to sub-sampling design choices, we conducted a simulation study varying both the number of sampled comparable pairs per

Normal errors Gumbel errors Logistic errors

Figure 4: Training loss across epochs for the Linear mean function under different error distributions, with $n_{\text{train}} = 1000$ and 25% censoring.

uncensored residual, denoted by s, and the mini-batch size b. Specifically, we considered $s \in \{5, 10, 15\}$ and $b \in \{32, 64, 128\}$, while keeping all other modeling components fixed, including a nonlinear mean function with interactions, a normal error distribution, a single hidden layer, $\sigma_b = 0.5$, and a censoring rate of 25%. The learning rate was set to $\alpha = 10^{-4}$, and model performance was evaluated after 2, 3, 5, and 10 training epochs.

The objective of this analysis was to quantify the impact of sub-sampling parameters on both predictive accuracy and training stability. Model performance was assessed on an independent test set using the IPCW C-index to evaluate discriminative ability and the AMSE to assess calibration.

Table 4: Sensitivity analysis of sub-sampling configurations. Test performance is reported as IPCW C-index / AMSE for different numbers of sampled pairs s, mini-batch sizes b, training sample sizes, and training epochs.

s	b	$n_{\text{train}} = 1000$		$n_{\text{train}} = 5000$	
		Epoch 5	Epoch 10	Epoch 2	Epoch 3
5	32	0.851/9.50	0.880/7.43	0.880/6.45	0.895/6.65
	64	0.836/14.69	0.856/8.93	0.848/7.68	0.873/6.61
	128	0.821/19.08	0.840/14.39	0.830/12.69	0.843/8.92
10	32	0.886/7.57	0.917/7.208	0.9123/6.10	0.934/6.01
	64	0.856/8.77	0.894/7.38	0.886/6.51	0.910/6.02
	128	0.840/14.20	0.864/8.43	0.857/7.35	0.881/6.56
15	32	0.901/7.67	0.938/6.48	0.934/6.28	0.941/8.80
	64	0.882/7.67	0.914/7.61	0.908/6.25	0.933/6.18
	128	0.850/10.24	0.886/7.61	0.883/6.58	0.904/6.27

Table 4 reports the results for each combination of (s, b), with entries presented in the form IPCW C-index / AMSE across different numbers of training epochs. This comparison

highlights how increasing the number of sampled pairs per residual generally improves discrimination at the cost of higher computational burden, while larger mini-batch sizes tend to stabilize training but may slow convergence.

4.2.4 High-Dimensional Covariates Analysis

We evaluated the performance of the RNN-AFT model in high-dimensional settings, where the number of covariates p exceeds the number of signal features. Only the first five covariates contributed to the true mean function

$$f(\mathbf{X}_i) = 2x_{i1} - x_{i2} + 0.5x_{i3},$$

while the remaining covariates acted as pure noise. Covariates were generated as

$$x_{i1} \sim \text{Bernoulli}(0.5), \quad x_{ij} \sim \mathcal{N}\Big(x_{i,j-1}/3, 1\Big), \quad j = 2, \ldots, p,$$

introducing moderate correlation among features.

The training sample size was fixed at $n_{\text{train}} = 1000$, with $K_i \sim \text{Poisson}(\lambda = 2)$ gap times per subject. The test set contained $n_{\text{test}} = 2000$, Gaussian errors were used, and the censoring proportion was set to 25%. To assess robustness, we gradually increased the number of noise covariates from 0 to 100, generating the mean function as

$$f(\mathbf{x}) = x_1 - 6x_2 + 4x_3 + \sum_{k=4}^{K} \beta_k x_k, \quad \beta_k = 0,$$

so that the signal-to-noise ratio decreases with increasing p. Conventional AFT models are not expected to leverage irrelevant covariates, while deep neural networks can identify signal in high-dimensional spaces.

Figures 5–6c collectively illustrate RNN-AFT performance under high-dimensional scenarios. While the final AMSE and C-index remain strong for moderate p, test performance degrades for very high p, as evidenced by the train-test gaps (Figures 6a and 6b). The loss trajectories (Figure 6c) indicate that training remains stable and convergent, demonstrating that observed performance degradation is driven by statistical generalization limits rather than optimization failure. Overall, the RNN-AFT model is robust to high-dimensional noise covariates, maintaining accurate prediction and ranking where conventional models may fail.

4.3 Real Data Applications

4.3.1 The Datasets

We evaluated the proposed RNN-AFT model on two real-world recurrent event datasets to demonstrate its ability to capture temporal dependence and patient heterogeneity.

The first dataset examines repeated infections among **128** patients with Chronic Granulomatous Disease (CDG), publicly available from the `survival` R package (Therneau, 2020). Patients were followed from randomization until study end or dropout. The outcome of interest is the time to serious CGD infections, with each patient experiencing 0 to 7 events.

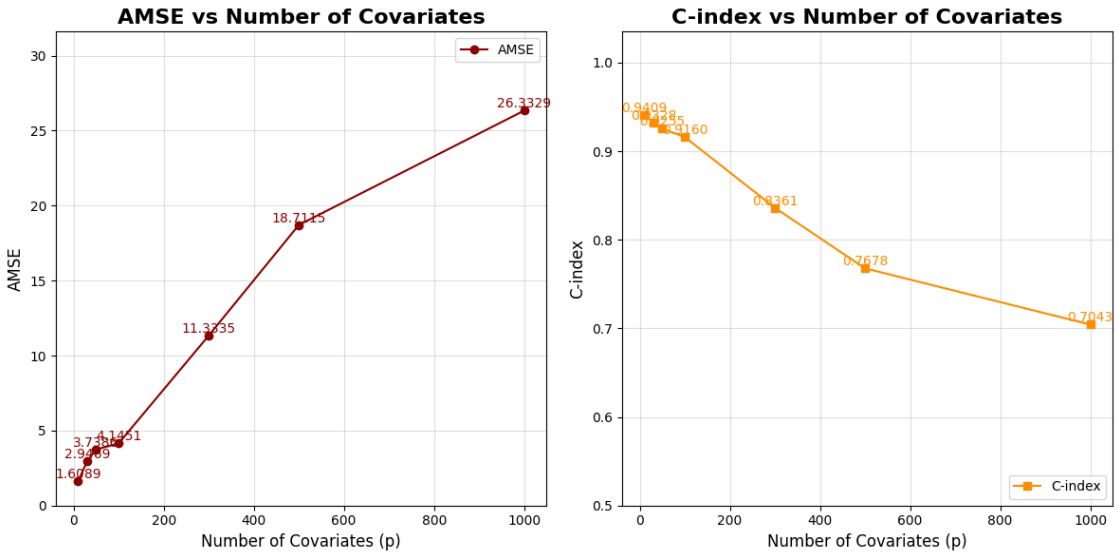


Figure 5: AMSE (left) and IPCW C-index (right) for the RNN-AFT model with increasing numbers of noise covariates.

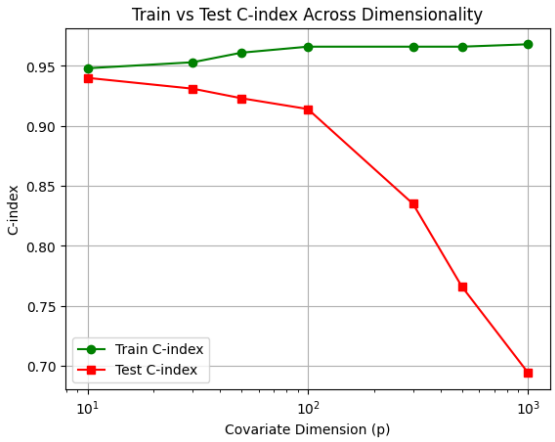
Time intervals are recorded using `tstart` and `tstop`, with `status` indicating occurrence of infection, and additional covariates include `id`, `sex`, `age`, and `treat` (placebo vs. gamma interferon). The median follow-up time was 293 days, and the primary aim is to assess the effect of gamma interferon on the incidence of repeated infections.

The second dataset consists of **403** colorectal cancer (CRC) patients following surgical treatment, originating from Gonzalez et al. (2005) and available through the `frailtypack` R package. Hospital readmissions were recorded as recurrent events, with right censoring occurring when patients were lost to follow-up or did not experience additional readmissions before study termination. Three time-fixed covariates were included: chemotherapy status (binary, 1 = received chemotherapy, 0 = no chemotherapy), Dukes' stage (1 = A–B, 2 = C, 3 = D), and sex (0 = male, 1 = female).

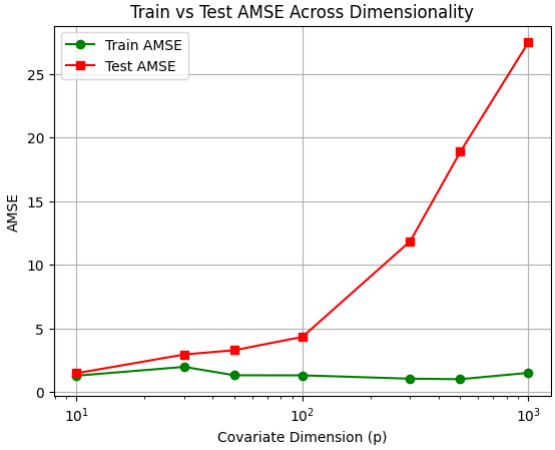
4.3.2 Predictive Performance

For both datasets, the RNN-AFT model was implemented using a gated recurrent unit (GRU) architecture with two recurrent layers and a hidden dimension of 64. Right censoring was handled via the pairwise mini-batch loss function with parameters $s = 30$ and $b = 64$. Model optimization used the RMSprop optimizer with a learning rate of 3×10^{-4}, and training was performed over 10 epochs on a CPU. Data were split into training (70%) and testing (30%) sets using a fixed random seed. Predictive accuracy was evaluated with the AMSE, and discriminative ability was assessed with the IPCW C-index, which accounts for censored observations.

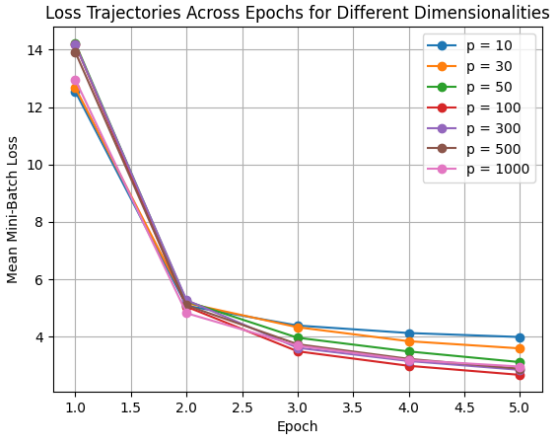
The results indicate that the RNN-AFT model achieves reasonable pointwise prediction (AMSE) and moderate discriminative ability (IPCW C-index) despite the small to moderate sample sizes and presence of censoring. The use of identical optimizer settings for both datasets demonstrates the model's robustness and stable generalization, effectively capturing



(a) Train vs Test C-index



(b) Train vs Test AMSE



(c) Loss trajectories across epochs

Figure 6: (a) Train vs Test C-index, (b) Train vs Test AMSE, and (c) Loss trajectories across epochs for different covariate dimensions. The train-test gaps increase for high p, while loss convergence remains stable across all dimensions.

Table 5: Predictive performance of the RNN-AFT model on two biomedical recurrent event datasets, evaluated using IPCW C-index and AMSE metrics.

Dataset	Train		Test	
	C-index	AMSE	C-index	AMSE
CGD Study (128 pts)	0.525	41.24	0.631	28.31
CRC Study (403 pts)	0.618	63.31	0.637	42.95

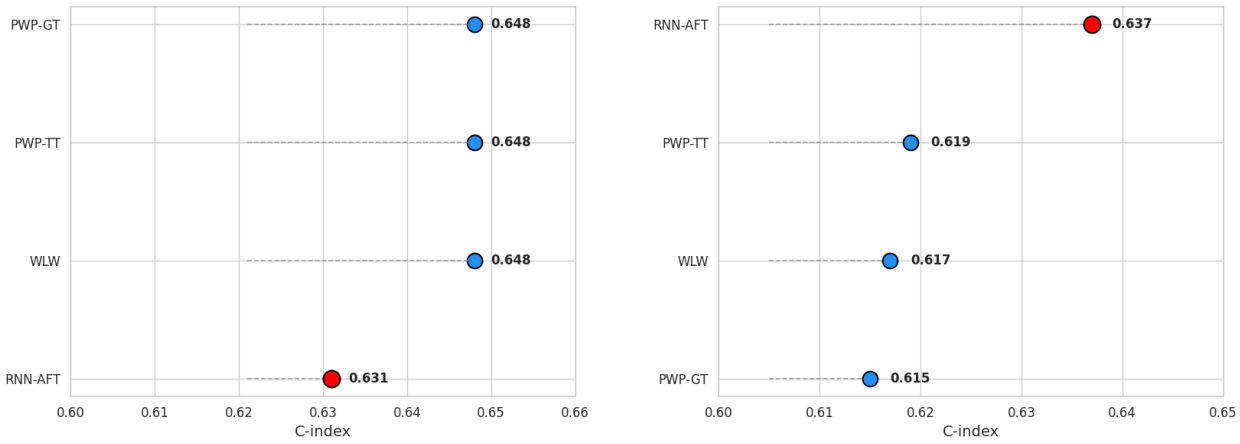
recurrent event dynamics and covariate effects for clinical risk stratification and dynamic outcome prediction.

4.3.3 Benchmarking Against Cox-based Models

To benchmark the RNN-AFT model against classical recurrent event approaches, we compared its predictive discrimination to gap-time–based Prentice–Williams–Peterson (PWP) models and the Wei–Lin–Weissfeld (WLW) model using the C-index for both datasets.

For the CGD dataset with 128 patients, the RNN-AFT achieved a testing C-index of 0.631. In comparison, classical Cox-based models performed slightly better than random baseline but were very close to each other: WLW (0.648), PWP-TT (0.648), and PWP-GT (0.648). While these models handle recurrent events differently—WLW models marginal hazards for each event order, PWP-TT focuses on total times conditional on previous events, and PWP-GT examines successive gap times—they all rely on a Cox-type multiplicative structure with a linear predictor, limiting their ability to capture nonlinear or history-dependent effects. The RNN-AFT effectively models complex temporal dependencies and nonlinear relationships, achieving competitive performance.

For the CRC dataset with 403 patients, the RNN-AFT achieved a testing C-index of 0.637, outperforming PWP-GT (0.615), PWP-TT (0.619), and WLW (0.617). Similar to the CGD study, classical Cox-based approaches often struggle to fully leverage nonlinear and patient-history information, whereas the RNN-AFT captures these effects effectively.



(a) CGD Dataset

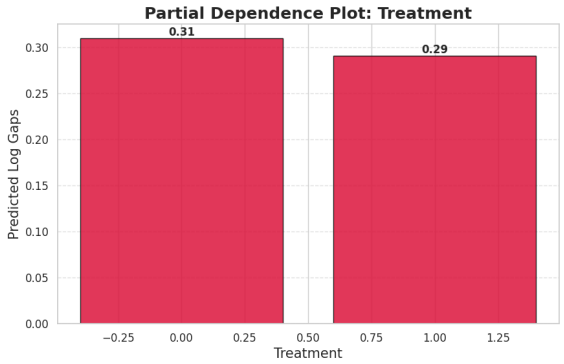
(b) CRC Dataset

Figure 7: Predictive performance comparison of RNN-AFT, PWP-GT, PWP-TT, and WLW models measured by IPCW C-index across two biomedical recurrent event datasets.

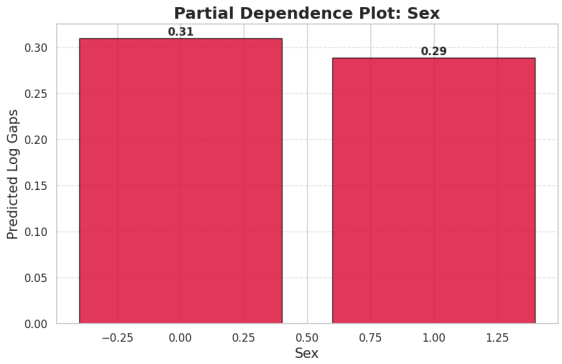
4.3.4 Sensitivity Analysis

To enhance interpretability of the proposed RNN-AFT model, we examine partial dependence plots (PDPs) for key time-fixed covariates in both datasets. PDP analysis isolates the marginal effect of each covariate on the predicted log gap time while averaging over the em-

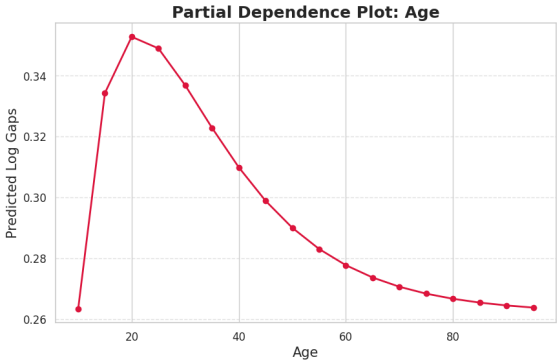
pirical distribution of the remaining variables, providing insight into how individual patient characteristics influence recurrent event dynamics within the learned nonlinear model.



(a) Treatment (Gamma Interferon)



(b) Sex



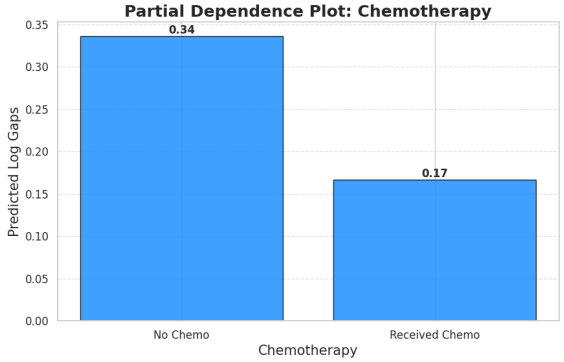
(c) Age at Randomization

Figure 8: Partial dependence plots for the Chronic Granulomatous Disease dataset illustrating the marginal effect of treatment, sex, and age on predicted log gap times between infections.

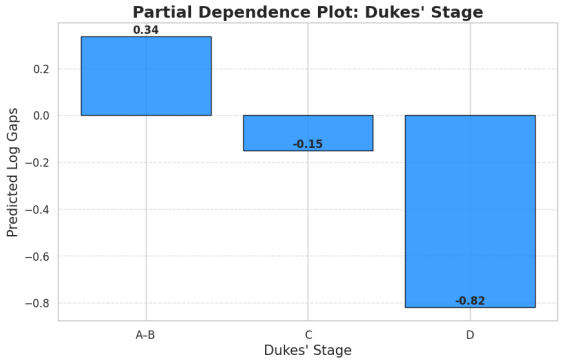
For the CGD study, patients receiving gamma interferon exhibit predicted gap times between infections that are comparable to those receiving placebo, suggesting no clear protective effect of treatment in this setting. Sex does not appear to have a meaningful influence on predicted infection intervals. In contrast, age shows a pronounced nonlinear effect: predicted gap times increase rapidly up to approximately 20 years of age and then gradually decrease thereafter, indicating a higher infection risk among older patients.

For the CRC study, patients receiving chemotherapy tend to have shorter predicted log gap times, suggesting a higher short-term risk of hospital readmission. Female patients exhibit longer predicted gap times than males, indicating a protective effect of sex. Higher Dukes' stages are associated with shorter predicted gaps, reflecting more frequent readmissions for advanced disease.

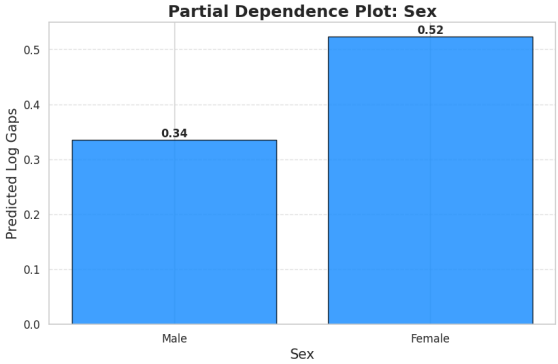
These analyses demonstrate that the RNN-AFT model effectively captures nonlinear covariate effects and temporal dependencies in both datasets. The PDPs highlight clinically meaningful trends and support interpretability of model predictions for recurrent event outcomes.



(a) Chemotherapy status



(b) Dukes' stage



(c) Sex

Figure 9: Partial dependence plots for the Colorectal Cancer dataset showing the marginal effect chemotherapy status, Dukes' stage and sex on predicted log gap times between hospital readmissions.

5 Discussion

The RNN-AFT model demonstrates strong performance in modeling recurrent event data, effectively capturing nonlinear effects of time-fixed covariates while handling right-censoring and maintaining high concordance between predicted and observed event times. Our training approach balances pairwise loss optimization with absolute log-gap prediction, highlighting a trade-off between accurately ranking events and minimizing the average mean squared error (AMSE) of predictions. In practice, careful selection of the sub-sample size m for pairwise updates, coupled with early stopping criteria, can improve computational efficiency and prevent overfitting; however, larger sub-sample sizes increase both memory and runtime requirements. Despite these strengths, the current model does not explicitly accommodate time-dependent covariates or very long-range correlations between recurrent events, and excessive sub-sample sizes can exacerbate overfitting. Future work could explore adaptive or importance sampling strategies for pair selection and hybrid loss functions that combine rank-based and regression-based objectives to further improve prediction accuracy and robustness.